Data Collection and Cleaning

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#Initial Cleaning

Load Packages

```
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tibble)
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.2

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(zoo)
```

```
## Warning: package 'zoo' was built under R version 4.5.2
```

```
##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

Read Raw Data

```r
games <- read_csv("Games.csv", show_col_types = FALSE)
opp_pg <- read_csv("Opponent Stats Per Game.csv", show_col_types = FALSE)
```

Inspect Column Names

```r
names(games)
```

```
##  [1] "gameId"           "gameDateTimeEst"  "hometeamCity"     "hometeamName"
##  [5] "hometeamId"       "awayteamCity"     "awayteamName"     "awayteamId"
##  [9] "homeScore"        "awayScore"        "winner"           "gameType"
## [13] "gameSubtype"      "gameLabel"        "gameSubLabel"     "seriesGameNumber"
## [17] "attendance"       "arenaId"          "arenaName"        "arenaCity"
## [21] "arenaState"       "officials"
```

```r
names(opp_pg)
```

```
##  [1] "season"            "lg"                "team"
##  [4] "abbreviation"      "playoffs"          "g"
##  [7] "mp_per_game"       "opp_fg_per_game"   "opp_fga_per_game"
## [10] "opp_fg_percent"    "opp_x3p_per_game"  "opp_x3pa_per_game"
## [13] "opp_x3p_percent"   "opp_x2p_per_game"  "opp_x2pa_per_game"
## [16] "opp_x2p_percent"   "opp_ft_per_game"   "opp_fta_per_game"
## [19] "opp_ft_percent"    "opp_orb_per_game"  "opp_drb_per_game"
## [22] "opp_trb_per_game"  "opp_ast_per_game"  "opp_stl_per_game"
## [25] "opp_blk_per_game"  "opp_tov_per_game"  "opp_pf_per_game"
## [28] "opp_pts_per_game"
```

Historical Base Dataset

```r
games_master <- games %>%
  select(
    game_id   = gameId,
    date      = gameDateTimeEst,
    home_team = hometeamName,
    away_team = awayteamName,
    home_pts  = homeScore,
    away_pts  = awayScore
```

```r
  ) %>%
  mutate(
    date   = as.Date(date),
    season = if_else(month(date) >= 10, year(date) + 1L, year(date)),
    spread = home_pts - away_pts,
    total  = home_pts + away_pts
  )
```

Team-Level Opponent Stats Dataset

```r
opp_team <- opp_pg %>%
  select(
    season,
    team,
    opp_fg_per_game,
    opp_fga_per_game,
    opp_fg_percent,
    opp_x3p_per_game,
    opp_x3pa_per_game,
    opp_x3p_percent,
    opp_x2p_per_game,
    opp_x2pa_per_game,
    opp_x2p_percent,
    opp_ft_per_game,
    opp_fta_per_game,
    opp_ft_percent,
    opp_orb_per_game,
    opp_drb_per_game,
    opp_trb_per_game,
    opp_ast_per_game,
    opp_stl_per_game,
    opp_blk_per_game,
    opp_tov_per_game,
    opp_pf_per_game,
    opp_pts_per_game
  ) %>%
  distinct()
```

Check Team Name Formats

```r
unique(games_master$home_team)[1:30]
```

```
##  [1] "Kings"         "Lakers"        "Timberwolves"  "Bulls"
##  [5] "76ers"         "Raptors"       "Heat"          "Hornets"
##  [9] "Cavaliers"     "Magic"         "Warriors"      "Jazz"
## [13] "Bucks"         "Wizards"       "Clippers"      "Celtics"
## [17] "Mavericks"     "Hawks"         "Pacers"        "Nets"
## [21] "Nuggets"       "Knicks"        "Thunder"       "Pistons"
## [25] "Suns"          "Rockets"       "Grizzlies"     "Trail Blazers"
## [29] "Pelicans"      "Spurs"
```

```r
unique(opp_team$team)[1:30]
```

```
##  [1] "Atlanta Hawks"          "Boston Celtics"         "Brooklyn Nets"
##  [4] "Chicago Bulls"          "Charlotte Hornets"      "Cleveland Cavaliers"
##  [7] "Dallas Mavericks"       "Denver Nuggets"         "Detroit Pistons"
## [10] "Golden State Warriors"  "Houston Rockets"        "Indiana Pacers"
## [13] "Los Angeles Clippers"   "Los Angeles Lakers"     "Memphis Grizzlies"
## [16] "Miami Heat"             "Milwaukee Bucks"        "Minnesota Timberwolves"
## [19] "New Orleans Pelicans"   "New York Knicks"        "Oklahoma City Thunder"
## [22] "Orlando Magic"          "Philadelphia 76ers"     "Phoenix Suns"
## [25] "Portland Trail Blazers" "Sacramento Kings"       "San Antonio Spurs"
## [28] "Toronto Raptors"        "Utah Jazz"              "Washington Wizards"
```

```r
sort(unique(games_master$season))[1:10]
```

```
##  [1] 1947 1948 1949 1950 1951 1952 1953 1954 1955 1956
```

```r
sort(unique(opp_team$season))[1:10]
```

```
##  [1] 1947 1948 1949 1950 1951 1952 1953 1954 1955 1956
```

Team Name Table

```r
team_lookup <- tibble(
  short_team = c(
    "Hawks", "Celtics", "Nets", "Hornets", "Bulls", "Cavaliers",
    "Mavericks", "Nuggets", "Pistons", "Warriors", "Rockets", "Pacers",
    "Clippers", "Lakers", "Grizzlies", "Heat", "Bucks", "Timberwolves",
    "Pelicans", "Knicks", "Thunder", "Magic", "76ers", "Suns",
    "Trail Blazers", "Kings", "Spurs", "Raptors", "Jazz", "Wizards"
  ),
  full_team = c(
    "Atlanta Hawks", "Boston Celtics", "Brooklyn Nets", "Charlotte Hornets", "Chicago Bulls", "Clevelan
    "Dallas Mavericks", "Denver Nuggets", "Detroit Pistons", "Golden State Warriors", "Houston Rockets"
    "Los Angeles Clippers", "Los Angeles Lakers", "Memphis Grizzlies", "Miami Heat", "Milwaukee Bucks",
    "New Orleans Pelicans", "New York Knicks", "Oklahoma City Thunder", "Orlando Magic", "Philadelphia
    "Portland Trail Blazers", "Sacramento Kings", "San Antonio Spurs", "Toronto Raptors", "Utah Jazz",
  )
)
```

Map Names

```r
games_master <- games_master %>%
  left_join(team_lookup, by = c("home_team" = "short_team")) %>%
  rename(home_team_full = full_team) %>%
  left_join(team_lookup, by = c("away_team" = "short_team")) %>%
  rename(away_team_full = full_team)
```

Preparing Opponet Datasets

```r
home_opp <- opp_team %>%
  rename_with(~ paste0("home_", .x), -c(season, team))

away_opp <- opp_team %>%
  rename_with(~ paste0("away_", .x), -c(season, team))
```

Joining Opponet Stats onto Master Dataset

```r
games_master <- games_master %>%
  left_join(home_opp, by = c("season", "home_team_full" = "team")) %>%
  left_join(away_opp, by = c("season", "away_team_full" = "team"))
```

Creating Difference Variables

```r
games_master <- games_master %>%
  mutate(
    opp_pts_allowed_diff = home_opp_pts_per_game - away_opp_pts_per_game,
    opp_orb_allowed_diff = home_opp_orb_per_game - away_opp_orb_per_game,
    opp_trb_allowed_diff = home_opp_trb_per_game - away_opp_trb_per_game,
    opp_ast_allowed_diff = home_opp_ast_per_game - away_opp_ast_per_game,
    opp_tov_forced_diff  = home_opp_tov_per_game - away_opp_tov_per_game
  )
```

Filtering To Modern Era

```r
games_master_modern <- games_master %>%
  filter(season >= 2010)
```

Previewing Current Dataset

```r
games_master_modern %>%
  select(
    game_id, date, season, home_team, home_team_full, away_team, away_team_full,
    home_pts, away_pts, spread, total,
    home_opp_pts_per_game, away_opp_pts_per_game,
    opp_pts_allowed_diff, opp_orb_allowed_diff,
    opp_trb_allowed_diff, opp_ast_allowed_diff, opp_tov_forced_diff
  ) %>%
  head(20)
```

```
## # A tibble: 20 x 18
##     game_id date       season home_team  home_team_full away_team away_team_full
##       <dbl> <date>      <dbl> <chr>      <chr>          <chr>     <chr>
##  1 22500892 2026-03-03   2026 Kings      Sacramento Ki~ Suns      Phoenix Suns
##  2 22500891 2026-03-03   2026 Lakers     Los Angeles L~ Pelicans  New Orleans P~
##  3 22500890 2026-03-03   2026 Timberwol~ Minnesota Tim~ Grizzlies Memphis Grizz~
##  4 22500889 2026-03-03   2026 Bulls      Chicago Bulls  Thunder   Oklahoma City~
##  5 22500888 2026-03-03   2026 76ers      Philadelphia ~ Spurs     San Antonio S~
##  6 22500887 2026-03-03   2026 Raptors    Toronto Rapto~ Knicks    New York Knic~
##  7 22500886 2026-03-03   2026 Heat       Miami Heat     Nets      Brooklyn Nets
##  8 22500883 2026-03-03   2026 Hornets    Charlotte Hor~ Mavericks Dallas Maveri~
```

```
##  9 22500884 2026-03-03   2026 Cavaliers  Cleveland Cav~ Pistons   Detroit Pisto~
## 10 22500885 2026-03-03   2026 Magic      Orlando Magic  Wizards   Washington Wi~
## 11 22500882 2026-03-02   2026 Warriors   Golden State ~ Clippers  Los Angeles C~
## 12 22500881 2026-03-02   2026 Jazz       Utah Jazz      Nuggets   Denver Nuggets
## 13 22500880 2026-03-02   2026 Bucks      Milwaukee Buc~ Celtics   Boston Celtics
## 14 22500879 2026-03-02   2026 Wizards    Washington Wi~ Rockets   Houston Rocke~
## 15 22500878 2026-03-01   2026 Lakers     Los Angeles L~ Kings     Sacramento Ki~
## 16 22500877 2026-03-01   2026 Clippers   Los Angeles C~ Pelicans  New Orleans P~
## 17 22500874 2026-03-01   2026 Celtics    Boston Celtics 76ers     Philadelphia ~
## 18 22500876 2026-03-01   2026 Mavericks  Dallas Maveri~ Thunder   Oklahoma City~
## 19 22500873 2026-03-01   2026 Hawks      Atlanta Hawks  Trail Bl~ Portland Trai~
## 20 22500875 2026-03-01   2026 Magic      Orlando Magic  Pistons   Detroit Pisto~
## # i 11 more variables: home_pts <dbl>, away_pts <dbl>, spread <dbl>,
## #   total <dbl>, home_opp_pts_per_game <dbl>, away_opp_pts_per_game <dbl>,
## #   opp_pts_allowed_diff <dbl>, opp_orb_allowed_diff <dbl>,
## #   opp_trb_allowed_diff <dbl>, opp_ast_allowed_diff <dbl>,
## #   opp_tov_forced_diff <dbl>
```

```r
write_csv(games_master, "master_dataset_v1.csv")
write_csv(games_master_modern, "master_dataset_modern_v1.csv")
```

#Creating OREB

```r
team_stats <- read_csv("TeamStatistics.csv", show_col_types = FALSE)
```

```r
names(team_stats)
```

```
##  [1] "gameId"                  "gameDateTimeEst"
##  [3] "teamCity"                "teamName"
##  [5] "teamId"                  "opponentTeamCity"
##  [7] "opponentTeamName"        "opponentTeamId"
##  [9] "home"                    "win"
## [11] "teamScore"               "opponentScore"
## [13] "assists"                 "blocks"
## [15] "steals"                  "fieldGoalsAttempted"
## [17] "fieldGoalsMade"          "fieldGoalsPercentage"
## [19] "threePointersAttempted"  "threePointersMade"
## [21] "threePointersPercentage" "freeThrowsAttempted"
## [23] "freeThrowsMade"          "freeThrowsPercentage"
## [25] "reboundsDefensive"       "reboundsOffensive"
## [27] "reboundsTotal"           "foulsPersonal"
## [29] "turnovers"               "plusMinusPoints"
## [31] "numMinutes"              "q1Points"
## [33] "q2Points"                "q3Points"
## [35] "q4Points"                "benchPoints"
## [37] "biggestLead"             "biggestScoringRun"
## [39] "leadChanges"             "pointsFastBreak"
## [41] "pointsFromTurnovers"     "pointsInThePaint"
## [43] "pointsSecondChance"      "timesTied"
## [45] "timeoutsRemaining"       "seasonWins"
## [47] "seasonLosses"            "coachId"
```

Building Offensive Rebound Dataset

```r
oreb_game <- team_stats %>%
  select(
    game_id = gameId,
    team = teamName,
    home = home,
    oreb = reboundsOffensive
  )
```

Splitting Into Home and Away

```r
# Home Offensive Rebounds

home_oreb <- oreb_game %>%
  filter(home == 1) %>%
  select(
    game_id,
    home_team = team,
    home_oreb = oreb
  ) %>%
  mutate(game_id = as.character(game_id))

away_oreb <- oreb_game %>%
  filter(home == 0) %>%
  select(
    game_id,
    away_team = team,
    away_oreb = oreb
  ) %>%
  mutate(game_id = as.character(game_id))
```

Joining Onto Master

```r
## Join Offensive Rebounds onto Master Dataset

games_master <- games_master %>%
  select(-any_of(c("home_oreb", "away_oreb", "oreb"))) %>%
  mutate(game_id = as.character(game_id)) %>%
  left_join(home_oreb, by = c("game_id", "home_team")) %>%
  left_join(away_oreb, by = c("game_id", "away_team")) %>%
  mutate(
    oreb = home_oreb + away_oreb
  )
```

Verifying

```r
games_master %>%
  select(
    game_id,
    home_team,
    away_team,
    home_oreb,
    away_oreb,
```

```r
    oreb
  ) %>%
  head(20)
```

```
## # A tibble: 20 x 6
##    game_id  home_team    away_team     home_oreb away_oreb  oreb
##    <chr>    <chr>        <chr>             <dbl>     <dbl> <dbl>
##  1 22500892 Kings        Suns                 13        18    31
##  2 22500891 Lakers       Pelicans             11        10    21
##  3 22500890 Timberwolves Grizzlies            13         5    18
##  4 22500889 Bulls        Thunder              10        12    22
##  5 22500888 76ers        Spurs                 9         6    15
##  6 22500887 Raptors      Knicks                5         6    11
##  7 22500886 Heat         Nets                 14        10    24
##  8 22500883 Hornets      Mavericks            13         7    20
##  9 22500884 Cavaliers    Pistons              11        10    21
## 10 22500885 Magic        Wizards               3        13    16
## 11 22500882 Warriors     Clippers             15        15    30
## 12 22500881 Jazz         Nuggets               8         8    16
## 13 22500880 Bucks        Celtics               9        19    28
## 14 22500879 Wizards      Rockets               7        21    28
## 15 22500878 Lakers       Kings                 9        16    25
## 16 22500877 Clippers     Pelicans             10        14    24
## 17 22500874 Celtics      76ers                19        10    29
## 18 22500876 Mavericks    Thunder               9         7    16
## 19 22500873 Hawks        Trail Blazers        13         9    22
## 20 22500875 Magic        Pistons               9        14    23
```

Rebuilding Modern

```r
games_master_modern <- games_master %>%
  filter(season >= 2010)
```

Save

```r
write_csv(games_master, "master_dataset_v2.csv")
```

```r
write_csv(games_master_modern, "master_dataset_modern_v2.csv")
```

#Adding Advanced Team Stats

Advanced Team File

```r
team_adv <- read_csv("TeamStatisticsAdvanced.csv", show_col_types = FALSE)

names(team_adv)
```

```
##  [1] "gameId"           "teamId"           "teamCity"         "teamName"
##  [5] "gameType"         "astPct"           "astRatio"         "astTo"
##  [9] "availableFlag"    "defRating"        "drebPct"          "eDefRating"
## [13] "eNetRating"       "eOffRating"       "ePace"            "efgPct"
## [17] "gameDate"         "gameDateTimeEst"  "home"             "matchup"
```

```
## [21] "min"              "netRating"        "offRating"        "opponentTeamCity"
## [25] "opponentTeamId"   "opponentTeamName" "orebPct"          "pace"
## [29] "pacePer40"        "pie"              "poss"             "rebPct"
## [33] "teamAbbreviation" "teamName_right"   "tmTovPct"         "tsPct"
## [37] "win"              "wl"
```

```r
head(team_adv)
```

```
## # A tibble: 6 x 38
##     gameId teamId teamCity teamName gameType astPct astRatio astTo availableFlag
##      <dbl>  <dbl> <chr>    <chr>    <chr>     <dbl>    <dbl> <dbl>         <dbl>
## 1 22500742 1.61e9 Detroit  Pistons  Regular~  0.705     23.3  2.38             1
## 2 22500745 1.61e9 Portland Trail B~ Regular~  0.633     20.9  1.72             1
## 3 22500744 1.61e9 New Orl~ Pelicans Regular~  0.674     21.2  1.81             1
## 4 22500746 1.61e9 LA       Clippers Regular~  0.595     16.9  1.38             1
## 5 22500743 1.61e9 Indiana  Pacers   Regular~  0.743     20.6  2.89             1
## 6 22500741 1.61e9 Boston   Celtics  Regular~  0.486     13.4  1.06             1
## # i 29 more variables: defRating <dbl>, drebPct <dbl>, eDefRating <dbl>,
## #   eNetRating <dbl>, eOffRating <dbl>, ePace <dbl>, efgPct <dbl>,
## #   gameDate <dttm>, gameDateTimeEst <chr>, home <dbl>, matchup <chr>,
## #   min <dbl>, netRating <dbl>, offRating <dbl>, opponentTeamCity <chr>,
## #   opponentTeamId <dbl>, opponentTeamName <chr>, orebPct <dbl>, pace <dbl>,
## #   pacePer40 <dbl>, pie <dbl>, poss <dbl>, rebPct <dbl>,
## #   teamAbbreviation <chr>, teamName_right <lgl>, tmTovPct <dbl>, ...
```

Column-Matcher

```r
## Helper to find likely column names

pick_col <- function(df, candidates) {
  nms <- names(df)
  hit <- nms[tolower(nms) %in% tolower(candidates)]
  if (length(hit) == 0) return(NA_character_)
  hit[1]
}
```

Identify Important Columns

```r
## Identify advanced-stat columns

game_id_col <- pick_col(team_adv, c("gameId"))
home_col    <- pick_col(team_adv, c("home"))

off_col  <- pick_col(team_adv, c("offensiveRating", "offRating", "off_rtg"))
def_col  <- pick_col(team_adv, c("defensiveRating", "defRating", "def_rtg"))
net_col  <- pick_col(team_adv, c("netRating", "net_rtg"))
pace_col <- pick_col(team_adv, c("pace"))

game_id_col
```

```
## [1] "gameId"
```

```r
home_col
```

```
## [1] "home"
```

```r
off_col
```

```
## [1] "offRating"
```

```r
def_col
```

```
## [1] "defRating"
```

```r
net_col
```

```
## [1] "netRating"
```

```r
pace_col
```

```
## [1] "pace"
```

Game_ids cleaning

```r
games_master <- games_master %>%
  mutate(game_id = as.character(game_id))

team_adv_small <- team_adv %>%
  transmute(
    game_id = as.character(.data[[game_id_col]]),
    home    = .data[[home_col]],
    off_rtg = .data[[off_col]],
    def_rtg = .data[[def_col]],
    net_rtg = .data[[net_col]],
    pace    = .data[[pace_col]]
  )
```

Splitting Home vs. Away

```r
home_adv <- team_adv_small %>%
  filter(home == 1) %>%
  select(
    game_id,
    home_off_rtg = off_rtg,
    home_def_rtg = def_rtg,
    home_net_rtg = net_rtg,
    home_pace    = pace
  ) %>%
  distinct(game_id, .keep_all = TRUE)

away_adv <- team_adv_small %>%
```

```r
  filter(home == 0) %>%
  select(
    game_id,
    away_off_rtg = off_rtg,
    away_def_rtg = def_rtg,
    away_net_rtg = net_rtg,
    away_pace    = pace
  ) %>%
  distinct(game_id, .keep_all = TRUE)
```

Rejoining

```r
games_master <- games_master %>%
  left_join(home_adv, by = "game_id") %>%
  left_join(away_adv, by = "game_id") %>%
  mutate(
    off_rtg_diff = home_off_rtg - away_off_rtg,
    def_rtg_diff = home_def_rtg - away_def_rtg,
    net_rtg_diff = home_net_rtg - away_net_rtg,
    pace_diff    = home_pace - away_pace
  )
```

IDs Check

```r
class(games_master$game_id)
```

```
## [1] "character"
```

```r
class(team_adv_small$game_id)
```

```
## [1] "character"
```

```r
length(intersect(games_master$game_id, team_adv_small$game_id))
```

```
## [1] 4995
```

```r
sum(games_master$game_id %in% team_adv_small$game_id)
```

```
## [1] 4995
```

viewing master

```r
games_master %>%
  filter(!is.na(home_off_rtg)) %>%
  select(
    game_id, home_team, away_team,
    home_off_rtg, away_off_rtg, off_rtg_diff,
    home_def_rtg, away_def_rtg, def_rtg_diff,
    home_net_rtg, away_net_rtg, net_rtg_diff,
    home_pace, away_pace, pace_diff
  ) %>%
  head(20)
```

```
## # A tibble: 20 x 15
##    game_id  home_team     away_team    home_off_rtg away_off_rtg off_rtg_diff
##    <chr>    <chr>         <chr>               <dbl>        <dbl>        <dbl>
##  1 22500745 Trail Blazers Grizzlies           130.         111.        19.2
##  2 22500746 Kings         Clippers            114.         115.        -0.800
##  3 22500744 Timberwolves  Pelicans            119.         123.        -4.10
##  4 22500743 Bucks         Pacers              114.         106.         7.60
##  5 22500742 Pistons       Knicks              127.          84.2       42.7
##  6 22500741 Celtics       Heat                106.         103.         3.30
##  7 22500739 Suns          Warriors            109          112.        -3.20
##  8 22500740 Lakers        76ers               114.         112.         2.70
##  9 22500738 Mavericks     Spurs               115          127.       -12.4
## 10 22500737 Rockets       Hornets             102.         114.       -11.4
## 11 22500736 Raptors       Bulls               131.         112.        19.4
## 12 22500735 Hawks         Jazz                122.         120.         2
## 13 22500734 Magic         Nets                122.         100         21.6
## 14 22500733 Pistons       Wizards             115.         124.        -8.8
## 15 22500732 Clippers      Cavaliers            91.9        125.       -33.4
## 16 22500731 Kings         Grizzlies           122.         125.        -2.70
## 17 22500730 Spurs         Thunder             110.         101          9.5
## 18 22500729 Bucks         Pelicans            133          130.         2.5
## 19 22500728 Rockets       Celtics             102.         127.       -24.5
## 20 22500727 Raptors       Timberwolves        122.         123.        -0.800
## # i 9 more variables: home_def_rtg <dbl>, away_def_rtg <dbl>,
## #   def_rtg_diff <dbl>, home_net_rtg <dbl>, away_net_rtg <dbl>,
## #   net_rtg_diff <dbl>, home_pace <dbl>, away_pace <dbl>, pace_diff <dbl>
```

#Adding Four Factors

Loading Dataset

```
team_ff <- read_csv("TeamStatisticsFourFactors.csv", show_col_types = FALSE)
```

Inspecting Columns

```
names(team_ff)
```

```
##  [1] "gameId"           "teamId"           "teamCity"         "teamName"
##  [5] "gameType"         "availableFlag"    "efgPct"           "ftaRate"
##  [9] "gameDate"         "gameDateTimeEst"  "home"             "matchup"
## [13] "min"              "oppEfgPct"        "oppFtaRate"       "oppOrebPct"
## [17] "oppTovPct"        "opponentTeamCity" "opponentTeamId"   "opponentTeamName"
## [21] "orebPct"          "teamAbbreviation" "teamName_right"   "tmTovPct"
## [25] "win"              "wl"
```

```
head(team_ff)
```

```
## # A tibble: 6 x 26
##     gameId     teamId teamCity   teamName  gameType availableFlag efgPct ftaRate
##      <dbl>      <dbl> <chr>      <chr>     <chr>            <dbl>  <dbl>   <dbl>
## 1 22500743 1610612749 Milwaukee  Bucks     Regular~             1  0.532   0.063
## 2 22500746 1610612758 Sacramento Kings     Regular~             1  0.5     0.239
```

```
## 3 22500742 1610612752 New York   Knicks    Regular~             1  0.402   0.207
## 4 22500742 1610612765 Detroit    Pistons   Regular~             1  0.633   0.181
## 5 22500741 1610612738 Boston     Celtics   Regular~             1  0.476   0.274
## 6 22500745 1610612757 Portland   Trail Bl~ Regular~             1  0.639   0.233
## # i 18 more variables: gameDate <dttm>, gameDateTimeEst <chr>, home <dbl>,
## #   matchup <chr>, min <dbl>, oppEfgPct <dbl>, oppFtaRate <dbl>,
## #   oppOrebPct <dbl>, oppTovPct <dbl>, opponentTeamCity <chr>,
## #   opponentTeamId <dbl>, opponentTeamName <chr>, orebPct <dbl>,
## #   teamAbbreviation <chr>, teamName_right <lgl>, tmTovPct <dbl>, win <dbl>,
## #   wl <chr>
```

Identifying Columns

```r
ff_game_id_col <- pick_col(team_ff, c("gameId"))
ff_home_col    <- pick_col(team_ff, c("home"))

efg_col <- pick_col(team_ff, c("efgPct"))
tov_col <- pick_col(team_ff, c("tmTOVPct"))
orb_col <- pick_col(team_ff, c("orebPct"))
ftr_col <- pick_col(team_ff, c("ftaRate"))

ff_game_id_col
```

```
## [1] "gameId"
```

```r
ff_home_col
```

```
## [1] "home"
```

```r
efg_col
```

```
## [1] "efgPct"
```

```r
tov_col
```

```
## [1] "tmTovPct"
```

```r
orb_col
```

```
## [1] "orebPct"
```

```r
ftr_col
```

```
## [1] "ftaRate"
```

Building Reduced Table

```r
team_ff_small <- team_ff %>%
  transmute(
    game_id = as.character(.data[[ff_game_id_col]]),
    home    = .data[[ff_home_col]],
    efg_pct = .data[[efg_col]],
    tov_pct = .data[[tov_col]],
    orb_pct = .data[[orb_col]],
    ft_rate = .data[[ftr_col]]
  )
```

Splitting Into Home and Away

```r
home_ff <- team_ff_small %>%
  filter(home == 1) %>%
  select(
    game_id,
    home_efg_pct = efg_pct,
    home_tov_pct = tov_pct,
    home_orb_pct = orb_pct,
    home_ft_rate = ft_rate
  ) %>%
  distinct(game_id, .keep_all = TRUE)

away_ff <- team_ff_small %>%
  filter(home == 0) %>%
  select(
    game_id,
    away_efg_pct = efg_pct,
    away_tov_pct = tov_pct,
    away_orb_pct = orb_pct,
    away_ft_rate = ft_rate
  ) %>%
  distinct(game_id, .keep_all = TRUE)
```

Join onto master

```r
games_master <- games_master %>%
  mutate(game_id = as.character(game_id)) %>%
  left_join(home_ff, by = "game_id") %>%
  left_join(away_ff, by = "game_id") %>%
  mutate(
    efg_pct_diff = home_efg_pct - away_efg_pct,
    tov_pct_diff = home_tov_pct - away_tov_pct,
    orb_pct_diff = home_orb_pct - away_orb_pct,
    ft_rate_diff = home_ft_rate - away_ft_rate
  )
```

Verifying

```r
games_master %>%
  filter(!is.na(home_efg_pct)) %>%
  select(
```

```r
    game_id, home_team, away_team,
    home_efg_pct, away_efg_pct, efg_pct_diff,
    home_tov_pct, away_tov_pct, tov_pct_diff,
    home_orb_pct, away_orb_pct, orb_pct_diff,
    home_ft_rate, away_ft_rate, ft_rate_diff
  ) %>%
  head(20)
```

```
## # A tibble: 20 x 15
##    game_id  home_team     away_team    home_efg_pct away_efg_pct efg_pct_diff
##    <chr>    <chr>         <chr>               <dbl>        <dbl>        <dbl>
##  1 22500745 Trail Blazers Grizzlies           0.639        0.563      0.0760
##  2 22500746 Kings         Clippers            0.5          0.585     -0.085
##  3 22500744 Timberwolves  Pelicans            0.489        0.631     -0.142
##  4 22500743 Bucks         Pacers              0.532        0.519      0.0130
##  5 22500742 Pistons       Knicks              0.633        0.402      0.231
##  6 22500741 Celtics       Heat                0.476        0.494     -0.0180
##  7 22500739 Suns          Warriors            0.561        0.566     -0.00500
##  8 22500740 Lakers        76ers               0.614        0.565      0.0490
##  9 22500738 Mavericks     Spurs               0.546        0.595     -0.0490
## 10 22500737 Rockets       Hornets             0.506        0.511     -0.00500
## 11 22500736 Raptors       Bulls               0.644        0.549      0.095
## 12 22500735 Hawks         Jazz                0.531        0.525      0.00600
## 13 22500734 Magic         Nets                0.591        0.488      0.103
## 14 22500733 Pistons       Wizards             0.517        0.637     -0.12
## 15 22500732 Clippers      Cavaliers           0.5          0.592     -0.092
## 16 22500731 Kings         Grizzlies           0.505        0.6       -0.095
## 17 22500730 Spurs         Thunder             0.515        0.5        0.0150
## 18 22500729 Bucks         Pelicans            0.723        0.606      0.117
## 19 22500728 Rockets       Celtics             0.482        0.539     -0.0570
## 20 22500727 Raptors       Timberwolves        0.612        0.588      0.0240
## # i 9 more variables: home_tov_pct <dbl>, away_tov_pct <dbl>,
## #   tov_pct_diff <dbl>, home_orb_pct <dbl>, away_orb_pct <dbl>,
## #   orb_pct_diff <dbl>, home_ft_rate <dbl>, away_ft_rate <dbl>,
## #   ft_rate_diff <dbl>
```

Checking For Overlap

```r
length(intersect(games_master$game_id, team_ff_small$game_id))
```

```
## [1] 4995
```

```r
sum(is.na(games_master$home_efg_pct))
```

```
## [1] 67883
```

```r
sum(is.na(games_master$away_efg_pct))
```

```
## [1] 67883
```

```r
sum(is.na(games_master$efg_pct_diff))
```

```
## [1] 67883
```

Filtering

```r
games_master_modern <- games_master %>%
  filter(!is.na(home_off_rtg)) %>%
  filter(!is.na(home_efg_pct))
```

Saving

```r
write_csv(games_master, "master_dataset_v4.csv")
write_csv(games_master_modern, "master_dataset_modern_v4.csv")
```

#Rolling Averages

Packages

```r
library(zoo)
```

Team-Level Game History

```r
team_games_roll_home <- games_master %>%
  select(
    game_id,
    date,
    team = home_team,
    pts = home_pts,
    pts_allowed = away_pts,
    oreb = home_oreb
  )

team_games_roll_away <- games_master %>%
  select(
    game_id,
    date,
    team = away_team,
    pts = away_pts,
    pts_allowed = home_pts,
    oreb = away_oreb
  )

team_games_roll <- bind_rows(team_games_roll_home, team_games_roll_away) %>%
  mutate(date = as.Date(date)) %>%
  arrange(team, date, game_id)
```

Computing Rolling Features

```r
team_games_roll <- team_games_roll %>%
  group_by(team) %>%
  arrange(date, game_id, .by_group = TRUE) %>%
  mutate(
    last5_pts = rollmean(lag(pts), 5, fill = NA, align = "right"),
    last5_pts_allowed = rollmean(lag(pts_allowed), 5, fill = NA, align = "right"),
    last5_oreb = rollmean(lag(oreb), 5, fill = NA, align = "right")
  ) %>%
  ungroup()
```

Home Rolling Stats

```r
home_recent <- team_games_roll %>%
  select(
    game_id,
    team,
    last5_pts,
    last5_pts_allowed,
    last5_oreb
  ) %>%
  rename(
    home_team = team,
    home_last5_pts = last5_pts,
    home_last5_pts_allowed = last5_pts_allowed,
    home_last5_oreb = last5_oreb
  )
```

Away Rolling Stats

```r
away_recent_roll <- team_games_roll %>%
  select(
    game_id,
    team,
    last5_pts,
    last5_pts_allowed,
    last5_oreb
  ) %>%
  rename(
    away_team = team,
    away_last5_pts = last5_pts,
    away_last5_pts_allowed = last5_pts_allowed,
    away_last5_oreb = last5_oreb
  )
```

Joining with Master

```r
games_master <- games_master %>%
  select(-any_of(c(
    "home_last5_pts", "away_last5_pts", "last5_pts_diff",
    "home_last5_pts_allowed", "away_last5_pts_allowed", "last5_pts_allowed_diff",
    "home_last5_oreb", "away_last5_oreb", "last5_oreb_diff"
  ))) %>%
```

```r
  left_join(home_recent, by = c("game_id", "home_team")) %>%
  left_join(away_recent_roll, by = c("game_id", "away_team")) %>%
  mutate(
    last5_pts_diff = home_last5_pts - away_last5_pts,
    last5_pts_allowed_diff = home_last5_pts_allowed - away_last5_pts_allowed,
    last5_oreb_diff = home_last5_oreb - away_last5_oreb
  )
```

Verifying

```r
games_master %>%
  select(
    game_id, date, home_team, away_team,
    home_last5_pts, away_last5_pts, last5_pts_diff,
    home_last5_pts_allowed, away_last5_pts_allowed, last5_pts_allowed_diff,
    home_last5_oreb, away_last5_oreb, last5_oreb_diff
  ) %>%
  filter(!is.na(home_last5_pts) | !is.na(away_last5_pts)) %>%
  head(20)
```

```
## # A tibble: 20 x 13
##    game_id  date       home_team    away_team     home_last5_pts away_last5_pts
##    <chr>    <date>     <chr>        <chr>                  <dbl>          <dbl>
##  1 22500892 2026-03-03 Kings        Suns                    115.           95.6
##  2 22500891 2026-03-03 Lakers       Pelicans                113           120
##  3 22500890 2026-03-03 Timberwolves Grizzlies               113           119
##  4 22500889 2026-03-03 Bulls        Thunder                 108           116
##  5 22500888 2026-03-03 76ers        Spurs                   121.          116.
##  6 22500887 2026-03-03 Raptors      Knicks                  116           110.
##  7 22500886 2026-03-03 Heat         Nets                    123.          108.
##  8 22500883 2026-03-03 Hornets      Mavericks               123           114
##  9 22500884 2026-03-03 Cavaliers    Pistons                 113.          116.
## 10 22500885 2026-03-03 Magic        Wizards                 106.          110.
## 11 22500882 2026-03-02 Warriors     Clippers                116.          114.
## 12 22500881 2026-03-02 Jazz         Nuggets                 112.          121.
## 13 22500880 2026-03-02 Bucks        Celtics                 107           111.
## 14 22500879 2026-03-02 Wizards      Rockets                 112.          115.
## 15 22500878 2026-03-01 Lakers       Kings                   112.          113.
## 16 22500877 2026-03-01 Clippers     Pelicans                108.          120.
## 17 22500874 2026-03-01 Celtics      76ers                   112.          122.
## 18 22500876 2026-03-01 Mavericks    Thunder                 119.          117
## 19 22500873 2026-03-01 Hawks        Trail Blazers           115.          106
## 20 22500875 2026-03-01 Magic        Pistons                 114           120.
## # i 7 more variables: last5_pts_diff <dbl>, home_last5_pts_allowed <dbl>,
## #   away_last5_pts_allowed <dbl>, last5_pts_allowed_diff <dbl>,
## #   home_last5_oreb <dbl>, away_last5_oreb <dbl>, last5_oreb_diff <dbl>
```

Rebuild Modern

```r
games_master_modern <- games_master %>%
  filter(!is.na(home_off_rtg)) %>%
  filter(!is.na(home_efg_pct))
```

Save

```
write_csv(games_master, "master_dataset_v5.csv")
write_csv(games_master_modern, "master_dataset_modern_v5.csv")
```

#Schedule Factors Building Team Level Schedule

```
team_games_rest_home <- games_master %>%
  select(
    game_id,
    date,
    team = home_team
  )

team_games_rest_away <- games_master %>%
  select(
    game_id,
    date,
    team = away_team
  )

team_games_rest <- bind_rows(team_games_rest_home, team_games_rest_away) %>%
  mutate(date = as.Date(date)) %>%
  arrange(team, date, game_id)
```

Days Since Last Game

```
team_games_rest <- team_games_rest %>%
  group_by(team) %>%
  arrange(date, .by_group = TRUE) %>%
  mutate(
    rest_days = as.numeric(date - lag(date))
  ) %>%
  ungroup()
```

Back-to-Back Indictator

```
team_games_rest <- team_games_rest %>%
  mutate(
    back_to_back = if_else(rest_days == 1, 1, 0)
  )
```

Split Home vs Away Rest

```
home_rest <- team_games_rest %>%
  rename(home_team = team) %>%
  select(
    game_id,
    home_team,
    home_rest_days = rest_days,
    home_back_to_back = back_to_back
  )
```

```r
away_rest <- team_games_rest %>%
  rename(away_team = team) %>%
  select(
    game_id,
    away_team,
    away_rest_days = rest_days,
    away_back_to_back = back_to_back
  )
```

Join Onto Master

```r
games_master <- games_master %>%
  left_join(home_rest, by = c("game_id", "home_team")) %>%
  left_join(away_rest, by = c("game_id", "away_team")) %>%
  mutate(
    rest_diff = home_rest_days - away_rest_days
  )
```

Rebuild Modern

```r
games_master_modern <- games_master %>%
  filter(!is.na(home_off_rtg)) %>%
  filter(!is.na(home_efg_pct))
```

Save

```r
write_csv(games_master, "master_dataset_v6.csv")
write_csv(games_master_modern, "master_dataset_modern_v6.csv")
```

Verify

```r
games_master %>%
  select(
    game_id,
    home_team,
    away_team,
    home_rest_days,
    away_rest_days,
    rest_diff,
    home_back_to_back,
    away_back_to_back
  ) %>%
  head(20)
```

```
## # A tibble: 20 x 8
##    game_id  home_team    away_team     home_rest_days away_rest_days rest_diff
##    <chr>    <chr>        <chr>                  <dbl>          <dbl>     <dbl>
##  1 22500892 Kings        Suns                       2              5        -3
##  2 22500891 Lakers       Pelicans                   2              2         0
##  3 22500890 Timberwolves Grizzlies                  2              2         0
##  4 22500889 Bulls        Thunder                    2              2         0
```

```
##  5 22500888 76ers        Spurs                     2              2         0
##  6 22500887 Raptors      Knicks                    3              2         1
##  7 22500886 Heat         Nets                      3              2         1
##  8 22500883 Hornets      Mavericks                 3              2         1
##  9 22500884 Cavaliers    Pistons                   2              2         0
## 10 22500885 Magic        Wizards                   2              1         1
## 11 22500882 Warriors     Clippers                  2              1         1
## 12 22500881 Jazz         Nuggets                   2              1         1
## 13 22500880 Bucks        Celtics                   1              1         0
## 14 22500879 Wizards      Rockets                   2              2         0
## 15 22500878 Lakers       Kings                     1              3        -2
## 16 22500877 Clippers     Pelicans                  3              1         2
## 17 22500874 Celtics      76ers                     2              3        -1
## 18 22500876 Mavericks    Thunder                   2              2         0
## 19 22500873 Hawks        Trail Blazers             3              1         2
## 20 22500875 Magic        Pistons                   3              2         1
## # i 2 more variables: home_back_to_back <dbl>, away_back_to_back <dbl>
```

#Rolling Team Strength

Team-Level Advanced History

```
team_adv_home <- games_master %>%
  select(
    game_id,
    date,
    team = home_team,
    off_rtg = home_off_rtg,
    def_rtg = home_def_rtg,
    net_rtg = home_net_rtg,
    pace = home_pace
  )

team_adv_away <- games_master %>%
  select(
    game_id,
    date,
    team = away_team,
    off_rtg = away_off_rtg,
    def_rtg = away_def_rtg,
    net_rtg = away_net_rtg,
    pace = away_pace
  )

team_adv_games <- bind_rows(team_adv_home, team_adv_away) %>%
  arrange(team, date)
```

Computing Rolling Averages

```
team_adv_games <- team_adv_games %>%
  group_by(team) %>%
  arrange(date, .by_group = TRUE) %>%
  mutate(
    last5_off_rtg = zoo::rollapply(lag(off_rtg), 5, mean, fill = NA, align = "right"),
```

```r
    last5_def_rtg = zoo::rollapply(lag(def_rtg), 5, mean, fill = NA, align = "right"),
    last5_net_rtg = zoo::rollapply(lag(net_rtg), 5, mean, fill = NA, align = "right"),
    last5_pace    = zoo::rollapply(lag(pace), 5, mean, fill = NA, align = "right")
  ) %>%
  ungroup()
```

Split Home vs. Away

```r
home_recent_adv <- team_adv_games %>%
  rename(home_team = team) %>%
  select(
    game_id,
    home_team,
    home_last5_off_rtg = last5_off_rtg,
    home_last5_def_rtg = last5_def_rtg,
    home_last5_net_rtg = last5_net_rtg,
    home_last5_pace = last5_pace
  )

away_recent_adv <- team_adv_games %>%
  rename(away_team = team) %>%
  select(
    game_id,
    away_team,
    away_last5_off_rtg = last5_off_rtg,
    away_last5_def_rtg = last5_def_rtg,
    away_last5_net_rtg = last5_net_rtg,
    away_last5_pace = last5_pace
  )
```

Join Onto Master

```r
games_master <- games_master %>%
  left_join(home_recent_adv, by = c("game_id", "home_team")) %>%
  left_join(away_recent_adv, by = c("game_id", "away_team")) %>%
  mutate(
    last5_off_rtg_diff = home_last5_off_rtg - away_last5_off_rtg,
    last5_def_rtg_diff = home_last5_def_rtg - away_last5_def_rtg,
    last5_net_rtg_diff = home_last5_net_rtg - away_last5_net_rtg,
    last5_pace_diff = home_last5_pace - away_last5_pace
  )
```

Rebuild Modern

```r
games_master_modern <- games_master %>%
  filter(!is.na(home_off_rtg)) %>%
  filter(!is.na(home_efg_pct))
```

Save

```r
write_csv(games_master, "master_dataset_v7.csv")
write_csv(games_master_modern, "master_dataset_modern_v7.csv")
```

##Fixes

Removing Preseason

```r
games_master <- games_master %>%
  filter(!(month(date) == 10 & day(date) < 18))
```

Only NBA

```r
nba_teams <- team_lookup$short_team

games_master <- games_master %>%
  filter(home_team %in% nba_teams,
         away_team %in% nba_teams)
```

Removing games w/o full history

```r
games_master_modern <- games_master %>%
  filter(season >= 2010) %>%
  filter(!is.na(home_last5_pts), !is.na(away_last5_pts))
```

Rebuilding Final Model Dataset

```r
model_data <- games_master_modern %>%
  select(
    game_id,
    date,
    home_team,
    away_team,

    spread,
    total,
    oreb,

    off_rtg_diff,
    def_rtg_diff,
    net_rtg_diff,
    pace_diff,

    efg_pct_diff,
    tov_pct_diff,
    orb_pct_diff,
    ft_rate_diff,

    opp_pts_allowed_diff,
    opp_orb_allowed_diff,
    opp_trb_allowed_diff,
    opp_ast_allowed_diff,
    opp_tov_forced_diff,
```

```r
    last5_pts_diff,
    last5_pts_allowed_diff,
    last5_oreb_diff,

    rest_diff,
    home_back_to_back,
    away_back_to_back
  ) %>%
  tidyr::replace_na(list(
    rest_diff = 0,
    home_back_to_back = 0,
    away_back_to_back = 0
  )) %>%
  drop_na(spread, total)
```

Final Save

```r
write_csv(model_data, "FINAL_CLEANED_SET.csv")
```

##Additional Fixes

Loading

```r
model_data_v2 <- read_csv("FINAL_CLEANED_SET.csv", show_col_types = FALSE)
```

Inspecting Month Distribution

```r
model_data_v2 %>%
  mutate(
    date = as.Date(date),
    month = month(date)
  ) %>%
  count(month)
```

```
## # A tibble: 12 x 2
##    month     n
##    <dbl> <int>
##  1     1  3800
##  2     2  2907
##  3     3  3499
##  4     4  2306
##  5     5   696
##  6     6   133
##  7     7    49
##  8     8   123
##  9     9    46
## 10    10  1168
## 11    11  3248
## 12    12  3413
```

Normal NBA Szn Window

```r
model_data_v2 <- model_data_v2 %>%
  mutate(date = as.Date(date)) %>%
  filter(month(date) %in% c(10, 11, 12, 1, 2, 3, 4))
```

Removing Weak Variables

```r
model_data_v2 <- model_data_v2 %>%
  select(
    -any_of(c(
      "off_rtg_diff",
      "def_rtg_diff",
      "net_rtg_diff",
      "pace_diff",
      "efg_pct_diff",
      "tov_pct_diff",
      "orb_pct_diff",
      "ft_rate_diff"
    ))
  ) %>%
  drop_na()
```

Verify

```r
dim(model_data_v2)
```

```
## [1] 19780    18
```

```r
colSums(is.na(model_data_v2))
```

```
##                game_id                    date               home_team
##                      0                       0                       0
##              away_team                  spread                   total
##                      0                       0                       0
##                   oreb   opp_pts_allowed_diff   opp_orb_allowed_diff
##                      0                       0                       0
##   opp_trb_allowed_diff   opp_ast_allowed_diff    opp_tov_forced_diff
##                      0                       0                       0
##         last5_pts_diff last5_pts_allowed_diff        last5_oreb_diff
##                      0                       0                       0
##              rest_diff      home_back_to_back      away_back_to_back
##                      0                       0                       0
```

```r
sum(duplicated(model_data_v2$game_id))
```

```
## [1] 0
```

```r
model_data_v2 %>%
  mutate(month = month(date)) %>%
  count(month)
```

```
## # A tibble: 7 x 2
##   month     n
##   <dbl> <int>
## 1     1  3688
## 2     2  2821
## 3     3  3383
## 4     4  2232
## 5    10  1144
## 6    11  3180
## 7    12  3332
```

```r
summary(model_data_v2$spread)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -56.000  -7.000   3.000   2.433  11.000  73.000
```

```r
summary(model_data_v2$total)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   134.0   197.0   213.0   213.5   229.0   351.0
```

```r
summary(model_data_v2$oreb)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    2.00   17.00   21.00   21.22   25.00   54.00
```

Save

```r
write_csv(model_data_v2, "FINAL_CLEANED_SET_v2.csv")
```